HCIA-AI V4.0 Certification Practice Exam

Complete Mock Exam Question Bank: 66 Questions & Verified Answers | Passing Score: 600/1000

Prepared by: Umed Jamal Nouri • @UmedJaMala

Question 1

TRUE OR FALSE • 16 Points

Speech processing is a general term for various processing technologies used to research the voicing process, statistical features of speech signals, speech recognition, machine-based speech synthesis, and speech perception.

A. True

B. False

Question 2

TRUE OR FALSE • 16 Points

Huawei's all-scenario AI solutions can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.

A. True

B. False

Question 3

TRUE OR FALSE • 16 Points

Typically, in a network that uses sigmoid, the gradient decreases to 0 within five layers, making the network difficult to train.

A. True

B. False

Question 4

TRUE OR FALSE • 16 Points

When you train a deep neural network with the softsign function, if the network is very deep, an increasing number of back-propagated gradients fall into the saturation area, making the gradient module smaller and finally close to 0, and the weight cannot be updated.

A. True

B. False

Question 5

TRUE OR FALSE • 16 Points

The tanh function can effectively solve the vanishing gradient problem.

A. True

B. False

Page Answers: Q1: True • Q2: True • Q3: True • Q4: True • Q5: False

Question 6

TRUE OR FALSE • 16 Points

L1 regularization is referred to as weight decay.

A. True

B. False

Question 7

TRUE OR FALSE • 16 Points

In a random search, each setting item is sampled from possible parameter values to find the optimal parameter subset.

A. True

B. False

Question 8

TRUE OR FALSE • 16 Points

Assuming a dataset contains the areas and prices of 21,613 housing units in a city, you can use a classification model to predict the prices of other housing units in the city.

A. True

B. False

Question 9

TRUE OR FALSE • 16 Points

The emergence of foundation models represents a shift from weak Al to general Al.

A. True

B. False

Question 10

TRUE OR FALSE • 16 Points

Batch normalization is a technique used in deep learning to optimize training. It reduces the dependence on initial weights and learning rates, resulting in improved model performance.

A. True

B. False

Page Answers: Q6: False • Q7: True • Q8: False • Q9: False • Q10: True

Question 11

TRUE OR FALSE • 16 Points

In regression tasks, the most commonly used loss function is cross entropy cost.

A. True

B. False

Question 12

TRUE OR FALSE • 16 Points

In data parallelism, the data of each training batch is split into N parts, forward propagation is performed across computing units, and the loss value calculated by each card is aggregated on the master node.

A. True

B. False

Question 13

TRUE OR FALSE • 16 Points

In pipeline parallelism, the parameters of each layer of a model are split into multiple parts and distributed to multiple computing units. After the forward computation is complete, the gradients are combined on the master computing unit in the original splitting mode.

A. True

B. False

Question 14

TRUE OR FALSE • 16 Points

Model fine-tuning is to adjust the model structure and the number of trainable parameters based on the pre-trained model, and to use more diverse data for training.

A. True

B. False

Question 15

TRUE OR FALSE • 16 Points

The core idea of adapter tuning is to add adapter modules to specific layers of a pre-trained model. These modules usually consist of two small neural networks: one for dimensionality reduction and the otherfor augmentation.

A. True

B. False

Page Answers: Q11: False • Q12: False • Q13: False • Q14: False • Q15: True

Question 16 SINGLE-ANSWER QUESTION • 16 Points

Without considering any regularization terms, the support vectors of a support vector machine (SVM) are composed of ( ).

- **A.** Points on the separating hyperplane
- **B.** The points farthest from the separating hyperplane
- **C.** The points closest to the separating hyperplane
- **D.** Points of a certain type

Question 17 SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values is continuously updated using the gradient descent method to minimize the loss function?

- **A.** Hyperparameters
- **B.** Feature value
- **C.** Number of samples
- **D.** Parameters

Question 18 SINGLE-ANSWER QUESTION • 16 Points

The ReLU function is commonly used in deep learning neural networks. Which of the following is the value range of this function?

- **A.** [0, +∞)
- **B.** [0, 1]
- **C.** [-1, 1]
- **D.** [-1, 0]

Question 19 SINGLE-ANSWER QUESTION • 16 Points

The sigmoid activation function is monotonic and continuous, has bounded outputs, and makes the network easy to converge. It was popular for a while, but when the network is deep, which one of these problems is sigmoid associated with?

- **A.** Gradient reduction
- **B.** Vanishing gradient
- **C.** XOR Problem
- **D.** Overfitting

Question 20 SINGLE-ANSWER QUESTION • 16 Points

Which of the following functions can be used to alleviate the vanishing gradient problem?

- **A.** Sigmoid
- **B.** Tanh
- **C.** Softsign
- **D.** ReLU

Page Answers: Q16: C • Q17: D • Q18: A • Q19: B • Q20: D

Question 21 SINGLE-ANSWER QUESTION • 16 Points

A manufacturer wants to produce virtual assistants for hospital use. For virtual assistants, which of the following technologies is used for voiceprint recognition?

- **A.** Speech recognition and processing
- **B.** Image recognition and processing
- **C.** Expert system and knowledge graph
- **D.** Image generation and enhancement

Question 22 SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a use case for natural language processing?

- **A.** Public opinion analysis
- **B.** Machine translation
- **C.** Text classification
- **D.** Image recognition

Question 23 SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a recurrent neural network?

- **A.** RNN
- **B.** LSTM
- **C.** CNN
- **D.** GRU

Question 24 SINGLE-ANSWER QUESTION • 16 Points

Within the broad landscape of large language models built on the Transformer architecture, which roadmap does GPT belong to?

- **A.** Encoder-Only
- **B.** Decoder-Only
- **C.** Encoder-Decoder
- **D.** Decoder-Encoder

Question 25 SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about different optimizers is false?

- **A.** RMSprop solves the problem of the Adagrad optimizer ending too early.
- **B.** Compared with RMSprop, Adagrad is more sensitive to gradient changes.
- **C.** Both Adagrad and RMSprop can set adaptive learning rate for parameters.
- **D.** Adam requires bias correction for initial iterations.

Page Answers: **Q21: A • Q22: D • Q23: C • Q24: B • Q25: B**

Question 26

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and distinguish between images of apples and those of pears. What is the name of this learning method?

A. Supervised learning
B. Unsupervised learning
C. Semi-supervised learning
D. Reinforcement learning

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is true about classification models and regression models in machine learning?

A. For regression problems, the output variables are discrete values. For classification problems, the output variables are continuous.
B. The most commonly used indicators for evaluating regression and classification problems are the accuracy and the recall rate.
C. Overfitting may occur in both regression and classification problems.
D. Logistic regression is a typical regression model.

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about support vector machines (SVMs) is false?

A. SVMs are classification models. Their basic model is the linear classifier that maximizes the width of the gap between the two categories in the feature space.
B. SVMs also have a kernel trick, which makes them non-linear classifiers.
C. In the case of linear inseparability, non-linear mapping algorithms are used to convert the linearly inseparable samples of low-dimensional input space into samples of high-dimensional feature space. In this way, samples become linearly separable.
D. SVMs only apply to linear classification.

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an Al deep learning framework?

A. TensorFlow
B. Scikit-learn
C. Pytorch
D. Caffe

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following are used in TensorFlow to describe the computation process?

A. Parameter
B. Session
C. Data flow
D. Tensor

Page Answers: Q26: A • Q27: C • Q28: D • Q29: B • Q30: C

Question 31 SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a pooling layer with a stride of 2, the size of the output map is ( )

A. 30 x 30
B. 16 x 16
C. 29 x 29
D. 23 x 23

Question 32 SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a characteristic of PyTorch?

A. Python first
B. Easy to debug
C. Extensive ecosystem
D. Default use of static maps

Question 33 SINGLE-ANSWER QUESTION • 16 Points

Which of the following is not an advanced AI application?

A. Knowledge graph
B. Quantum computing
C. SDN
D. Intelligent driving

Question 34 SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is true about the training of large models?

A. Foundation models cannot be directly deployed.
B. The pre-training of large models requires a large amount of high-quality data.
C. Unlabeled data used in the pre-training phase of large models must be used in the fine-tuning phase.
D. Large models have emergent abilities, and therefore they do not need to be aligned with human values

Question 35 SINGLE-ANSWER QUESTION • 16 Points

Tensor parallelism is a common method for parallel training of large models. Which of the following types of model data is split to different devices?

A. Gradient
B. Optimizer
C. Input data
D. Model parameters

Page Answers: **Q31: B** • **Q32: D** • **Q33: C** • **Q34: B** • **Q35: D**

Question 36 MULTIPLE-ANSWER QUESTION • 18 Points

What are the main schools of thought in AI?

- **A.** Symbolism
- **B.** Illusionism
- **C.** Actionism
- **D.** Connectionism

Question 37 MULTIPLE-ANSWER QUESTION • 18 Points

What are the layers used in Transformer to capture time dependencies within a sequence?

- **A.** Residual layer
- **B.** Self-attention layer
- **C.** Location code
- **D.** Feed-forward network layer

Question 38 MULTIPLE-ANSWER QUESTION • 18 Points

During the calculation of the attention mechanism, which of the following are required for similarity calculation to obtain the attention score?

- **A.** Query
- **B.** Weight
- **C.** Value
- **D.** Key

Question 39 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common evaluation metrics for classification tasks?

- **A.** Accuracy
- **B.** Precision
- **C.** Recall rate
- **D.** Mean squared error

Question 40 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hyperparameters may be defined during model training?

- **A.** Batch size
- **B.** Learning rate
- **C.** Optimizer type
- **D.** Loss value

Page Answers: Q36: ACD • Q37: BC • Q38: AD • Q39: ABC • Q40: ABC

Question 41 MULTIPLE-ANSWER QUESTION • 18 Points

Optimizers are often used to accelerate training of deep learning neural networks. Which of the following are deep learning optimizers?

A. Momentum
B. Adagrad
C. RMSprop
D. Adam

Question 42 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about some of the subfields of AI are true?

A. Computer vision is a science that studies how to make computers "see" things.
B. Speech processing is a general term for different speech processing methods and techniques, including the study of vocalization, the collection of statistics related to speech signals, speech recognition, speech synthesis, and speech perception.
C. Natural language processing studies how to use computer technology to understand and use human language.
D. Autonomous driving does not require speech processing or computer vision.

Question 43 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common activation functions of neural networks?

A. Dropout
B. Sigmoid
C. tanh
D. Leaky ReLU

Question 44 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following comprise the Adam optimizer?

A. Momentum
B. Adagrad
C. RMSprop
D. Nesterov

Question 45 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about datasets are true?

A. A dataset is a collection of data used in machine learning tasks. Each piece of data is called a sample.
B. Events or attributes that reflect the particular performance or nature of a sample are called features.
C. A training set is a dataset used in the training process, where each sample is referred to as a training sample.
D. Learning (or training) is the process of building a model from data.

Page Answers: **Q41: ABCD** • **Q42: ABC** • **Q43: BCD** • **Q44: AC** • **Q45: ABCD**

Question 46 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about data preprocessing are true?

A. Data cleansing is a process of filling in missing values, as well as detecting and eliminating noise and exceptions.

B. Data dimension reduction aims to simplify data attributes and avoid the curse of dimensionality.

C. Data standardization aims to reduce noise data and improve model accuracy by standardizing data.

D. Machine learning tasks generate results in the form of model outputs. Therefore, model training is more important than data preprocessing.

Question 47 MULTIPLE-ANSWER QUESTION • 18 Points

Data augmentation can improve model robustness and avoid overfitting. Which of the following are data augmentation methods?

A. Batch normalization

B. Image contrast change

C. Image noise adding

D. Random crop

Question 48 MULTIPLE-ANSWER QUESTION • 18 Points

The softmax function is frequently used in classification tasks. Which of the following statements about it are true?

A. The softmax function is also called the normalized exponential function.

B. The softmax regression model is an algorithm for solving binary classification regression problems.

C. The softmax function is the generalization of the binary classification function sigmoid.

D. The softmax function is usually used together with the cross-entropy loss function.

Question 49 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are characteristics of eager execution in TensorFlow 2.x?

A. High performance

B. Deployability

C. Intuitiveness

D. Flexibility

Question 50 MULTIPLE-ANSWER QUESTION • 18 Points

The process of developing AI applications includes ( ).

A. Data preprocessing

B. Modeling

C. Model training

D. Model deployment

Page Answers: **Q46: ABC • Q47: BCD • Q48: ACD • Q49: CD • Q50: ABCD**

Question 51 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about large models are true?

A. A large amount of data is required for training.

B. The model parameter scale is large.

C. Training data does not need to be cleansed because this helps models learn how to process error information.

D. The generalization capability of large models is stronger than that of small models.

Question 52 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following fields can large model technologies be applied to?

A. Autonomous driving

B. Robots

C. Translation

D. Finance

Question 53 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following may be included in the training dataset of GPT-3?

A. COCO2017

B. Common Crawl

C. Wikipedia

D. Books1

Question 54 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following fine-tuning methods do not change the Transformer block structure of a large model?

A. Adapter tuning

B. Prefix tuning

C. Prompt tuning

D. LoRA

Question 55 MULTIPLE-ANSWER QUESTION • 18 Points

The development of large models requires not only training and fine-tuning, but also evaluation. Which of the following are reasons for evaluating large models?

A. Improving the accuracy of model outputs

B. Evaluating the capabilities of large models

C. Providing reference for enterprises to select models

D. Aligning large models with human values

Page Answers: **Q51: ABD** • **Q52: ABCD** • **Q53: BCD** • **Q54: BC** • **Q55: BC**

Question 56 FILL-IN-THE-BLANK QUESTION • 16 Points

On a convolution layer of a convolutional neural network, if there are 128 3 x 3 convolution kernels and the size of the input feature map is 28 x 28 x 64, the depth of a convolution kernel is ____. (Enter a number)

Answer: _____

Question 57 FILL-IN-THE-BLANK QUESTION • 16 Points

CNN stands for ____.

Answer: _____

Question 58 FILL-IN-THE-BLANK QUESTION • 16 Points

The ____ algorithm is based on the idea that if most of a sample's k nearest neighbors in the feature space belong to the same class, the sample also belongs to that class. (Enter an abbreviation.)

Answer: _____

Question 59 FILL-IN-THE-BLANK QUESTION • 16 Points

In multi-class tasks, the sum of all output probability values of the Softmax function is ____. (Enter a number.)

Answer: _____

Question 60 FILL-IN-THE-BLANK QUESTION • 16 Points

In k-fold cross-validation, the existing data is divided into ____ groups.

Answer: _____

Page Answers: Q56: 64 • Q57: Convolutional Neural Network • Q58: KNN • Q59: 1 • Q60: k

Question 61 TRUE OR FALSE • 16 Points

An AI speaker is a typical use case for speech processing.

A. True
B. False

Question 62 TRUE OR FALSE • 16 Points

If the hyperparameter search space is large, random search is better than grid search.

A. True
B. False

Question 63 SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is input and a 5 x 5 kernel is used for convolution with the stride of 1, the size of the output image is ( ).

A. 28 x 28
B. 29 x 29
C. 28 x 23
D. 23 x 23

Question 64 MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about model parameters and hyperparameters are true?

A. Models contain both parameters and hyperparameters.
B. Hyperparameters are automatically learned by models.
C. Hyperparameters are manually set.
D. Hyperparameters can be used to control training.

Question 65 FILL-IN-THE-BLANK QUESTION • 16 Points

The three essential elements for AI development are data, compute, and ____.

Answer: _____

Question 66 FILL-IN-THE-BLANK QUESTION • 16 Points

A/An ____ is a binary classification model that is based on a linear classifier defined in the feature space. (Enter an abbreviation.)

Answer: _____

Page Answers: Q61: True • Q62: True • Q63: A • Q64: ACD • Q65: Algorithm (or Algorithms) • Q66: SVM